Riphah International Colleges

A Project of Riphah International University

Course Outline

Course Information	**Course Title**	**Programming Fundamentals**		
	Course ID	CS-1113/CS 3133	**Course Type**	Major
	Credit hours	3 (2+1)	**Hours per week (C-L)**	3 (2+1)
	Program(s)	ADP IT Management	**Preferred Semester**	2nd
Course Description	This course offers an elementary grasp of the fundamentals of computer programming techniques and problem-solving techniques. Apart from general computing concepts it focuses on developing a problem-solving approach in the students and concentrates on formulation of algorithms and translating them into structured C++ programs. It includes the following major topics ♦ Introduction to C++ ♦ Data types and declaring and using variables ♦ Decision Control ♦ Repetitive Structures ♦ Functions, Call by Value, Call by reference, Recursion ♦ Arrays – Single and multi-dimensional arrays ♦ Passing arrays as an argument to the function.			
Course Objectives (CO)	*The objective of this course is to enable students to understand;*			
	No.	**Objective**		
	CO1.	To design algorithms to solve simple problem using tools like pseudocode, flow charts, and desk checking		
	CO2.	To implement algorithmic logic in programming language		
	CO3.	To correct, test and debug computer programs		
	CO4.	To explain how algorithms and computer programs work		
	CO5.	To demonstrate the basic structured programming in a team-oriented environment by conducting a term project		
Lecture type	**Class room Lectures, Lab Sessions, Project Presentation**			
Prerequisites	NIL			
Follow up Courses	Object Oriented Programming			

Text Book and Reference Books

	Title	Edition	Authors	Publisher	Year	ISBN
Textbook	Beginning C++20 From Novice to Professional	6th	Ivor Horton and Peter Van Weert	Apress	2020	978-1-4842-5883-5
Reference Books	♦ C++ How to Program Object Oriented Programming in C++	10th 4th	♦ D&D ♦ Robert Lafore	♦ Pearson ♦ The Waite's Group		♦ 0134448235 ♦ 0672323087
	C++ programming from problem analysis to program design	5th	DS Malik	CENGAGE		♦ 978-0-538-79808-2

Course Software or Tool	Microsoft Visual C++, Dev-C++, online Compilers or any other supporting Compilation Tool based Instructors' guideline

Grade Distribution:

Evaluation Type	Percentage (%)	Activities
Assignments & & Presentations	10%	Min. 4 in the semester
Quiz & Project	10%	Min. 4 in the semester
Lab	15%	Min. 1 (for Lab it is necessary)
Mid Term	25%	Contents from Week 1 to Week 8 will be included
Final Term	40%	Contents from Week 8 to Week 17
Total Points	100	
Methods of Evaluation	Quizzes, Assignments, Mid/Final exam, Lab, Project	
Notes	Labs are managed and evaluated separately	

Course Contents:

Week	Contents	Activities
01-02	**Introduction to Computers and Programming** • Why Program • Computer Systems: Hardware and Software • Programs and Programming Languages • What is Program Made of? • Input Processing and Output • The Programming Process • Procedural and Object-Oriented Programming	**Quiz-01** **(Week 2)**

03-04	**Introduction to C++** • The Parts of a C++ Program • The cout Object • The #include Directive • Variable, Literals and Assignment Statement • Identifiers • Integer Data Types • The char Data Types • The char Data Type • The C++ string Class • Floating-Point Data Types • The bool Data Type • Determining the Size of a Data Type • More about variable Assignments and Initialization • Scope • Arithmetic Operators • Comments	**Quiz-02** **(Week 4)**
	• Names Constants • Programming Style	
05-06	**Expression and Interactivity** • The cin Object • Mathematical Expressions • Type Conversion • Overflow and Underflow • Type Casting • Multiple Assignment and Combined Assignment • Formatting Output • Working with Characters and string Objects • More Mathematical Library Functions • Focus on Debugging: Hand Tracing a Prgram • Focus on Problem Solving; A Case Study	**Quiz-03** **(Week 6)** **Assignment 1** **(Week 6)**

07-08	**Making Decisions** • Relational Operators • The if Statement • Expanding the if Statement • The if /else Statement • Nested if Statements • The if/else if Statement • Flags • Logical Operators • Checking Numeric Ranges with Logical Operators • Menus • Focus on Software Engineering: Validating User Input • Comparing Characters and Strings • The Conditional Operator • The switch Statement • More about Blocks and Variable Scope	**Assignment 2** **(Week 7)**
MID TERM		
10-12	**Loops and Files** • The Increment and Decrement Operators • Introduction to Loops: The while Loop • Using the while Loop for Input Validation • Counters • The do-while Loop • The for Loop	**Quiz-4** **(Week 12)**

	• Keeping a Running Total • Sentinels • Focus on Software Engineering : Deciding Which Loop to Use • Nested Loops • Using Files for Data Storage • Optional Topics: Breaking and Continuing a Loop	**Assignment 3 (Week 12)**
13-14	**Functions** • Focus on Software Engineering: Modular Programming • Defining and Calling Functions • Function Prototypes • Sending Data into a Function • Passing Data by Value • Focus on Software Engineering: Using Functions in a Menu-Driven Program • The return Statement • Returning a Value from a Function • Returning a Boolean Value • Local and Global Variables • Static Local Variables • Default Arguments • Using Reference Variable as Parameters • Overloading Functions • The exit () Function • Stubs and Drivers	**Assignment 4 (Week 14)**
15-17	**Arrays and Vectors** • Arrays Hold Multiple Values • Accessing Array Elements • No Bounds Checking in C++ • The Range Based for Loop • Processing Array Contents • Focus on Software Engineering: Using Parallel Arrays • Arrays as Function Arguments • Two-Dimensional Arrays • Arrays with Three or More Dimensions • Focus on Problem Solving and Program Design: A Case Study\ • Introduction to the STL vector	**Project Presentation (week 16-17)**
FINAL TERM		